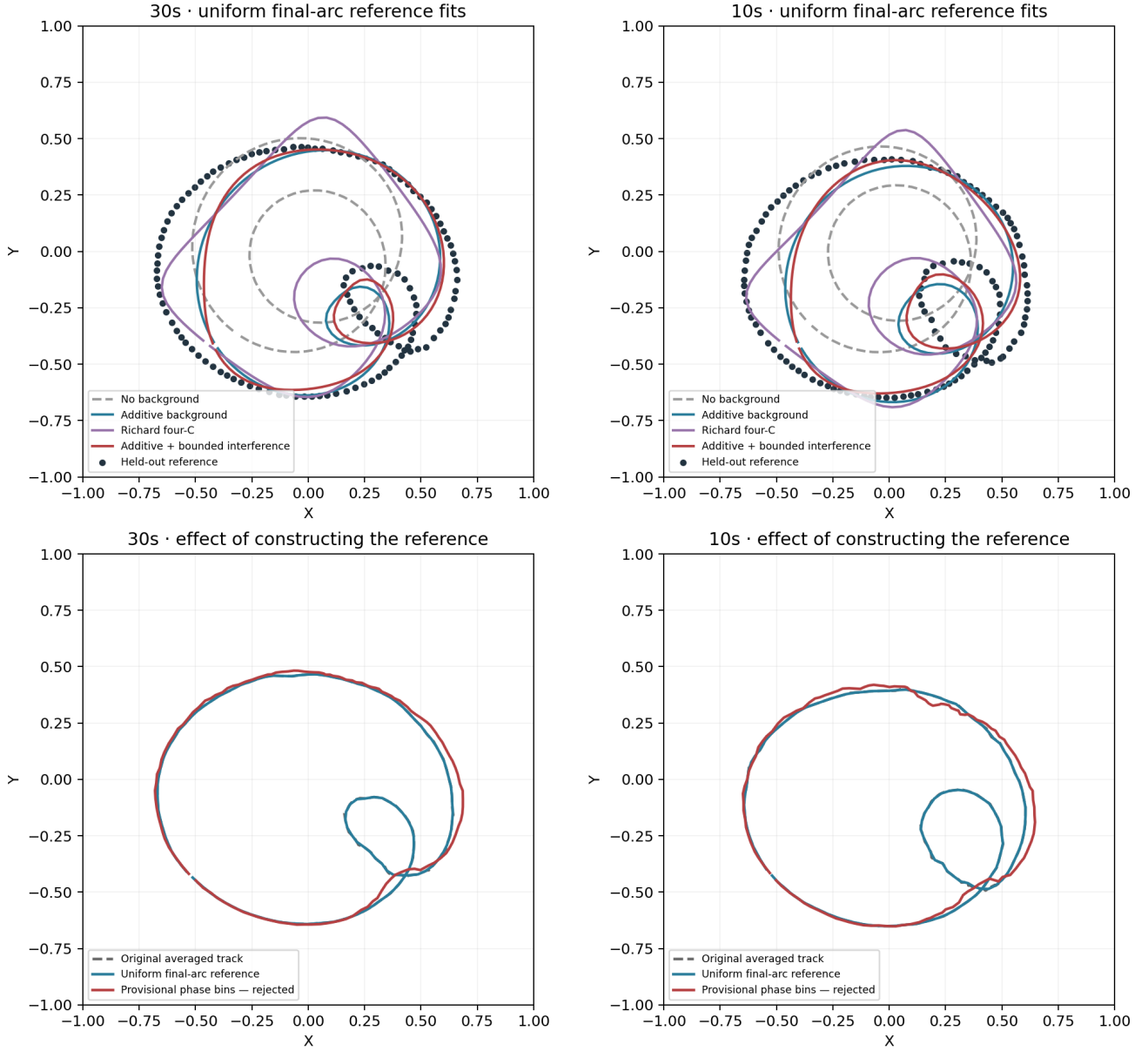


Reference distribution tests

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Result. Uniform arc resampling of the final averaged reference and arc weighting of the original points give similar physical fits. The simple direct-inversion sphere-circle guide does not provide a reliable monotone phase coordinate here: making it monotone largely removes the inner loop from the averaged reference. This is a reference-construction failure, not evidence that a physical background model fits better.



Primary comparison: weighting versus resampling

Two tests isolate the uneven-density concern. Weighted: preserve all original intensities and use trapezoidal arc weights. Uniform: interpolate the four channels together to 128 equally spaced locations along the final training-average arc; apply the identical training-derived interpolation map to the held-out average. Simulation is always matched using actual final-reference arc coordinates, one cyclic offset and both directions. No flexible phase warp.

$$J = \sum_k w_k \sum_i \left[\frac{I_{i,k}^{\text{pred}} - I_{i,k}^{\text{obs}}}{T_k^{\text{obs}}} \right]^2, \quad w_k \propto \frac{\Delta s_{k-1} + \Delta s_k}{2}.$$

For the uniform reference, bin weights are equal. To compare the two fitted models fairly, the table below evaluates both on the *same original held-out intensities with the same arc weights*. Their starting reference point is identical; no extra held-out alignment is fitted.

Window	Model	Weighted fit	Uniform fit	Uniform BG/ T_{max}	Uniform θ
30s	None	0.07633	0.07633	0.0%	30.3–46.8°
30s	Additive	0.02856	0.02857	35.5%	29.1–89.9°
30s	Richard four-C	0.03876	0.03876	—	29.0–88.6°
30s	Additive + interference	0.02612	0.02613	42.4%	30.0–90.0°
10s	None	0.07524	0.07526	0.0%	31.3–45.0°
10s	Additive	0.03217	0.03208	38.3%	33.7–89.9°
10s	Richard four-C	0.03883	0.03879	—	33.4–88.4°
10s	Additive + interference	0.02872	0.02860	39.8%	34.3–89.9°

Both preserve the measured two-loop trajectory. The residual mismatch remains after removing uneven-density weighting. This supports using the simpler uniform-final-arc reference for presentation and retaining arc weighting as a control. It does not prove the physical models or angles correct.

All four models retain fixed excitation with one brightness scale, original optics/calibration, and the same background budget in original intensity units. The additive budget is held fixed at 95% of the original training minimum across reference methods, rather than changing the budget when resampling changes a sampled minimum. Physical curves are not used to replace measured points. Mean-of-channel values are retained before calculating anisotropy.

Richard uses the stable equivalent parameterization $I_i = \alpha f_i + K_i \sqrt{f_i}$, with $\alpha = a_0^2$, $K_i = a_0 C_i$. Under these new arc-weight objectives the selected alpha values are finite (about 0.0192/0.00511 for uniform 30/10 s). The previously reported zero-brightness tendency therefore depends on weighting; the structural degeneracy of the unconstrained model has not been removed.

Nine base starts per model/window/reference cover both directions and four offsets, plus nested-model starts. Selected primary fits converge and have positive predicted channels. Local optimality is not a global guarantee. Uniformity refers to integrated arc on the reference interpolant; adjacent straight chord lengths need not be identical at curved sections.

Direct-inversion sphere-circle phase guide

Raw calibrated channels from 30–31 s and 10–11 s were checked: **zero samples outside the nominal inversion domain** in either 250,000-sample window. The user’s expectation about the range is supported on these data. A 41-sample channel smoother constructs the guide only; it does not replace the intensities being averaged.

For each ordered cycle, invert guide anisotropy to folded theta and unwrapped phi. Fit one provisional least-squares plane using equally weighted training cycles (128 time-decimated guide directions each). Project directions into that plane’s basis and unwrap their polar angle. This circle/plane sets a coordinate only; neither its cone geometry nor its projected points enter the physical fit as measured data.

The coordinate reverses substantially within individual traversals. Normalizing the first-to-last angle span to one traversal does not remove these reversals. A monotone least-squares (isotonic) guide was tried as a deliberately flagged sensitivity, while preserving acquisition order. Raw channel means in the resulting contiguous bins were then averaged with equal cycle weight. No cycles were removed (304 total, 152/152 split; 217 total, 109/108 split).

Guide diagnostic	30–31 s	10–11 s
Median maximum monotone adjustment	15.0%	21.7%
95th percentile maximum adjustment	28.3%	39.1%
Median adjustment with separate plane per cycle	15.4%	22.4%

These are fractions of each normalized provisional traversal, not angular measurement errors. The median raw plane-angle span is only about 0.53–0.54 turns per detected track. The uncorrected anisotropy winds once around the origin, so its half-angle inversion does not yield a closed directed upper-hemisphere trajectory. This is a limitation of this simple uncorrected guide, not an assertion that phase guidance can never work.

The guide is not one-to-one through the inner loop. Flattening its reversals concentrates different trajectory sections into a small phase region; averaging then largely erases the inner loop (bottom row of figure). This occurs even though every raw intensity sample is retained in some bin and no intensity is replaced by a circle prediction.

Disposition. Fits on the phase-binned references were completed as diagnostics, saved in fits.json/summary.csv, and explicitly excluded from the primary comparison. They should not be interpreted as corrected rod angles or improved fits. The 10 s complex phase-reference run also exhausted its refinement budget; no conclusion relies on its optimum. A separate best plane per cycle did not cure the guide-coordinate instability. No extra fitted phase flexibility or new inverse-circle constraints were introduced.

Files: prepare.py (raw reference construction), fit.py (24 model/reference/window fits), report.py, references.npz, preparation_checks.json, additional_guide_checks.json, fits.json and summary.csv. Reproduce using numpy/scipy/matplotlib/XeLaTeX builds this report. No commit or push.